How INLA works

Three easy steps for one county

Work in pairs. Keep a phone calculator handy.

The problem

Pine County had **9 deaths** where **5** were expected: SMR = $9/5 = 1.8$. That rests on very few deaths, so, like Restaurant A, Pine County should **borrow** from the rest of the state. **But how much?** Nobody knows. So INLA tries a few amounts and takes an average.

Step 1. Try three amounts of borrowing

4 minutes

Borrowing k means: add k deaths to the observed *and* k to the expected, then divide.

$$\text{best guess} = \frac{9+k}{5+k}$$

	Borrow k	Best guess
A borrow a little	5	$\frac{9+5}{5+5} = \frac{14}{10} = 1.40$
B borrow some	20	_____
C borrow a lot	100	_____

1. The more we borrow, the closer the best guess gets to _____, the state average.

Step 2. Average them

4 minutes

INLA does not pick one guess. It averages all three, giving more weight to the amount of borrowing that fits the data of **all the counties** best. R finds these weights for us:

	A a little	**B** some	**C** a lot
Weight	0.25	0.60	0.15

It works just like your **course grade**: homework 25%, exam 60%, quizzes 15%.

2. Pine County's final best guess:

$$0.25 \times 1.40 \; + \; 0.60 \times _____ \; + \; 0.15 \times _____ \; = \; _____$$

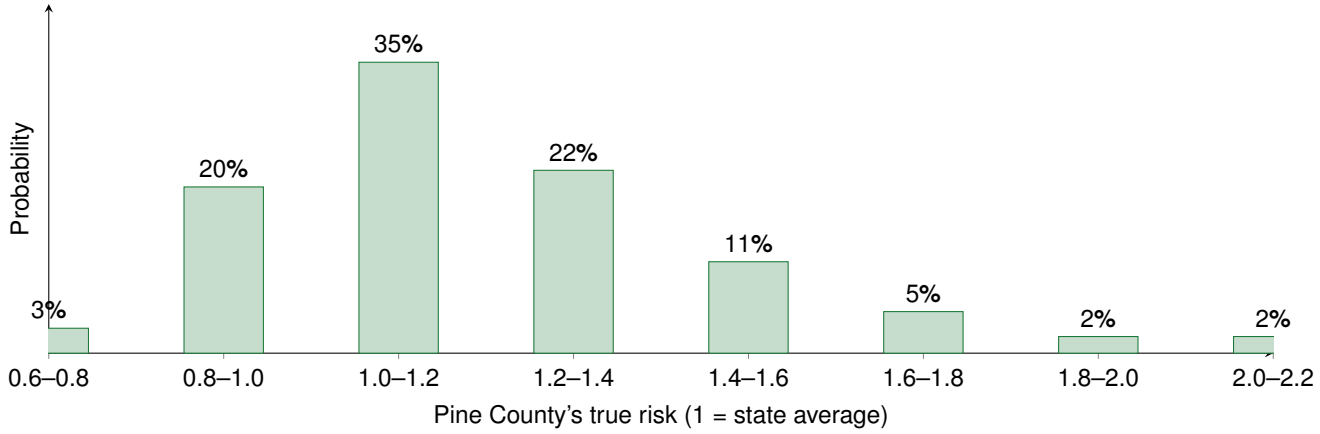
3. Why average the three instead of keeping only B?

☐ Because we are not sure how much to borrow.
☐ Because B is wrong.
☐ Because averaging always gives 1.

Step 3. Read the answer

6 minutes

INLA's final answer is not one number but a whole range of possible values, each with a probability. For Pine County:



4. How likely is Pine County above the state average? Add up the bars to the right of 1.0:

_____ + _____ + _____ + _____ + _____ + _____ = _____

This is the **exceedance probability**.

5. The 95% credible interval. Throw away the lowest 2.5% and the highest 2.5%. The lowest bar (3%) and the two highest bars (2% + 2%) hold just about that much, so the cut points are near the edges of those bars. Pine County's risk is between about **0.8** and _____, with 95% probability.

6. Pine County's SMR was 1.8. Is it clearly a high-risk county? Use questions 4 and 5.

In R

INLA does Steps 1 to 3 for every county at once. You read the results here:

On this handout	In R, after `m <- inla(...)`
best guess (Step 2)	`m$summary.fitted.values$mean`
95% credible interval (question 5)	`0.025quant` and `0.975quant` columns
probability above 1 (question 4)	`1 - inla.pmarginal(1, marg)`